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Key Points:

- Modeled hydraulic redistribution is compared and validated with field-measured sap flow data
- Water partitioning in the ecosystem with two vegetation species is tracked using a shared resource modeling
- Model quantifies the impact of hydraulic redistribution on transpiration and the competition for water between coexisting vegetation

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Impact of Hydraulic Redistribution on Multispecies Vegetation Water Use in a Semiarid Savanna Ecosystem: An Experimental and Modeling Synthesis

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Abstract A major challenge in critical zone science is to understand and predict the interaction between above-ground and below-ground ecohydrologic processes. One process that facilitates this connection is hydraulic redistribution, a phenomenon by which roots serve as preferential pathways for water movement from wet to dry soil layers. We use a multilayer canopy model in conjunction with experimental data to quantify the influence of hydraulic redistribution on ecohydrologic processes in order to characterize the competitive and facilitative interaction between mesquite trees and bunchgrasses in a semiarid savanna. Both measured and simulated results show that hydraulic descent dominates during the wet monsoon season, whereas hydraulic lift occurs between precipitation events. For 2015 year-long simulation, we find about 17% of precipitation is absorbed as soil moisture, with the rest of the precipitation returning to the atmosphere as evapotranspiration. In the wet season, 13% of precipitation is transferred to deep soil (>1.5 m) through roots, and in the dry season, 9% of this redistributed water is then transported back to shallow soil depths (<0.5 m). Assuming water supplied through hydraulic redistribution is well-mixed with moisture transported directly through the soil matrix and supports vegetation evapotranspiration, hydraulic redistribution supports 47% of mesquite transpiration and 9% of understory transpiration. Through modeling and experimental synthesis, this study demonstrates that in semiarid savanna ecosystems, mesquite exhibits a competitive advantage over understory bunchgrass through hydraulic redistribution. This analysis evaluates the relationship between two coexisting vegetation types that could be expanded to multiple vegetation species sharing resources in an ecosystem.

Plain Language Summary Water transported through roots, both upward and downward in the soil system, plays an important role in determining if coexisting vegetation species have facilitative or competitive dependences. Modeling and experimental results show that in the semiarid southwestern US, deep-rooted mesquite trees move water downward during and shortly after a rainfall event which then becomes available for transpiration during dry periods between rainfall events. This provides a competitive advantage for water use to mesquites over the understory bunchgrass. Through a combination of modeling and observational studies, we quantify the relative fractions of rainfall used in downward and upward water movements, also called hydraulic descent and hydraulic lift, respectively.

1. Introduction

The transport of water between plant roots and the soil affects above-ground and below-ground ecophysiological dynamics, energy, and nutrient exchange (Brooks et al., 2002, 2006; Burgess et al., 1998, 2000, 2001a, 2001b; Caldwell et al., 1998; Cardon et al., 2013; Caldwell & Richards, 1989; Dawson, 1993, 1996; Hultine et al., 2003, 2004; Neumann & Cardon, 2012; Richards & Caldwell, 1987; Schulze et al., 1998; Smith et al., 1999). Plant roots primarily uptake water and nutrients for metabolic activities. However, plant roots can also release water to the soil (Brooksbank et al., 2011; Kramer, 1993) and this phenomena has been

observed in different species across a wide range of climate zones, such as in semiarid deserts (Hultine et al., 2003; Ryel et al., 2002), savannas (Ludwig et al., 2003; Scott et al., 2008), temperate (Emerman & Dawson, 1996), and tropical forests (Meinzer et al., 2004; Oliveira et al., 2005). Hydraulic redistribution (HR) is used to describe this process by which water is distributed from wet to dry soil layers through roots that act as preferential flow paths (Burgess et al., 1998, 2000, 2001a; Hultine et al., 2003, 2004). HR, where water can move from near-surface soil to deeper layers or vice-versa, occurs as root hydraulic conductivity can be significantly higher than the surrounding soil hydraulic conductivity. Therefore, roots transport moisture at a higher rate than the soil does. Still, HR is passively driven by soil water potential gradients between shallow and deep soil layers, and between roots and the soil matrix. Plants may benefit from HR through enhanced net productivity and plant transpiration (TR) (Amenu & Kumar, 2008; Quijano et al., 2012; Scott et al., 2008), buffered soil moisture loss in the dry season (Bleby et al., 2010), and an extended growing season (Ryel et al., 2002, Scott et al., 2008). HR also supports plants by increasing deep nutrients mobility (McCulley et al., 2004), enhancing nutrient acquisition (Quijano et al., 2013), and facilitating root litter decomposition (McCulley et al., 2004).

HR may also serve to allow plants to alter water loss through evaporation and transpiration rates. For example, through hydraulic descent (HD), roots near the surface gather available water and move it toward the trunk, and then transports it downward, typically at nighttime or when the canopy is dormant. As a result, the amount of available water near the surface decreases and less water evaporates from surface soil. Through hydraulic lift (HL), plants transport water from deep to shallow soil layers, typically at night, which can support transpiration, either solely by the plant that lifted the water or the shallower rooted understory vegetation. Most studies related to HR have focused on its net impact on the transpiration and gross productivity of a single species of vegetation (Brooks et al., 2002; Caldwell & Richards, 1989; Ryel et al., 2002). Despite advances, we still know little about how might water utilization and movement by the overstory through HR impact near-surface available moisture also used by the understory, and how dynamic these patterns are across seasonally varying moisture regimes. Until recently, many of these above-ground and below-ground dynamics and overstory-understory interactions have been poorly constrained in models (David et al., 2013).

Whereas field measurements of water and carbon dynamics can identify the consequences of HR on ecosystem function, pairing the measurements with numerical modeling can provide novel insights into the underlying mechanisms that enable the dynamic water movement. The objective of this study is to understand the impact of HR on the amount of water that coexisting vegetation use for their transpiration by integrating modeling with field experimental data. In particular, we study the utilization of water in a semiarid savanna ecosystem using a “shared resource model” (Quijano et al., 2012), where overstory trees and understory grasses share the same soil moisture and nutrient resources and interact through the movement of soil moisture by vegetation roots. We assume that the soil moisture and hydraulically distributed water are well-mixed so that we can identify an ecosystem’s hydraulic redistribution regime (HRR)—an ecosystem metric that captures the balance in downward and upward movement of water (HL relative to HD), the seasonality of these bidirectional flows, and the resulting impacts on the plant communities (Barron-Gafford et al., 2017). This idea of HRR captures the dynamic heterogeneity of water availability that arises due to water uptake patterns of plant roots that mediate precipitation and groundwater regimes, such that an ecosystem’s HRR and climatic regime interact to determine understory and whole ecosystem functioning. Furthermore, we examine whether the presence of HR prompts competitive or facilitative dependencies between coexisting vegetation in water-limited environment (Ludwig et al., 2003; Quijano et al., 2012).

2. Study Site

Our study site is the Santa Rita Mesquite savanna (SRM), which is located on the Santa Rita Experimental Range (SRER) in southern Arizona (31.8214°N, 110.8661°W, elevation 1,116 m). This semiarid savanna lacks access to a stable groundwater source. The depth of groundwater at this site exceeds 100 m. The site lies in subtropical climate zone where 50% of annual rainfall occurs during summer North American Monsoon season (July–September) (Scott et al., 2008). The site has a distinct dry season and wet season that provides conditions for different types of HR (Scott et al., 2008). HL occurs during the dry season. During the wet season, mesquite roots transport water to the deep soil layer (HD) and the process is observed even during the

dormant canopy season (Hultine et al., 2004; Scott et al., 2008). Water stored in the deeper soil may benefit vegetation in the dry season and extend the growing season by providing them water in the absence of precipitation. Based on the relationship between mesquite sap flow and above-canopy fluxes, Scott et al. (2008) implied that mesquite stores water through HD to transpire more in the following spring. For the study site, field data is available for ecohydrologic model validation (Barron-Gafford et al., 2017) and can be used for studying the implications of HR on overstory-understory interactions and resulting spatial patterns and gradients.

The site has changed from semiarid grassland into a savanna due to the encroachment of the woody leguminous tree, *Prosopis velutina* woot (velvet mesquite) over the last 100 years (Glendening, 1952; McClaran, 2003). Currently, the vegetation of the site consists of mesquite trees and understory bunchgrasses that coexist and use the same water and nutrient resources. Average tree height is 2.5 m (SD = 1.6, $n = 95$). Mesquite has a deep taproot and wide-ranging lateral roots (Cable, 1977), which is suitable to redistribute below-ground water from different soil layers (Scott et al., 2008). Total mesquite cover is about 35%, understory cover is about 15%, with about half (7%) of the bunchgrass leaf area is attributable to plants growing directly under the mesquite canopy. Growth of mesquite and grass peaks during the middle of the summer monsoon (~August). During the dry season, understory plant cover decreases and patches of bare soil (loamy sand soils) occur between the perennial trees and bunchgrasses. Both mesquite and bunchgrass leaves senesce in winter and new leaves flush in spring. Green mesquite leaves endure through the dry period before the summer monsoon, whereas bunchgrasses die back and enter a dormant period.

3. Methods

3.1. Instrumentation and Ecosystem Monitoring

The SRM site is an Ameriflux study site that employs the eddy covariance technique used to measure land-atmosphere water, energy, and carbon dioxide fluxes (Scott et al., 2009). At 7.8 m height, an open-path infrared gas analyzer (LI-7500; LI-COR Inc., Lincoln, NE) and sonic anemometer (CSAT-3; Campbell Scientific Inc., Logan, UT) are installed to measure wind speed, air temperature, and densities of CO₂ and water vapor at 30 min interval. We used the measured data as atmospheric forcing for the multilayer canopy (MLCan) model (Drewry et al., 2010a, 2010b) and used the turbulent heat fluxes as validation of the model. Figure 1 shows that modeled energy fluxes provide a good estimate of measured energy fluxes. Most of the points in the scatter plot, in Figures 1d–1f, cluster along 1:1 line. Measured and modeled ground heat fluxes show close agreement (Figure 1d). The model slightly underestimates sensible heat and latent heat fluxes (Figures 1e and 1f).

Figure 2 shows the incoming shortwave radiation, precipitation, and leaf area indices (LAIs) of overstory mesquite and understory bunchgrasses. The seasonality of precipitation results in distinct wet and dry periods throughout the year. In combination with shortwave radiation, seasonal precipitation generates temporal soil water potential gradients and variation in HR fluxes. The LAIs of vegetation show how mesquite and bunchgrass respond seasonally. Total LAI was taken from MODIS (Oak Ridge National Laboratory Distributed Active Archive Center, 2016), which is available every 8 days at 1 km resolution. LAI was divided into two components, mesquite and bunchgrass, based on previous manual LAI measurements and personal observations made in the field. Accordingly, the increase in LAI in the early spring prior to tree leaf out, which occurs regularly each spring, was assumed to be bunchgrass LAI. Springtime bunchgrass LAI peaks prior to mesquite leaf out and goes to zero by early June, while mesquite LAI remains the same throughout the premonsoon period. During the monsoon, the increase in LAI was mainly attributed to bunchgrass with only a small, 0.1, increase in mesquite LAI, and bunchgrass LAI decays to zero by early November with any residual green LAI then attributed to mesquite. Partitioned LAI was scaled to the area of land occupied by vegetation, by dividing the LAI by the vegetation fraction (0.42, from transect data collected in 2014) and then fitted with a seven-order polynomial function for input into MLCan.

Also, volumetric soil content (CS616; Campbell Scientific, Logan, UT) and soil temperature (using thermocouples) were measured in 30 min intervals at 5, 10, 20, 30, 50, 70, 100, and 130 cm depth under a mesquite and in the intercanopy space. These data were used for further validation of the model (Figure A2 in Appendix A).

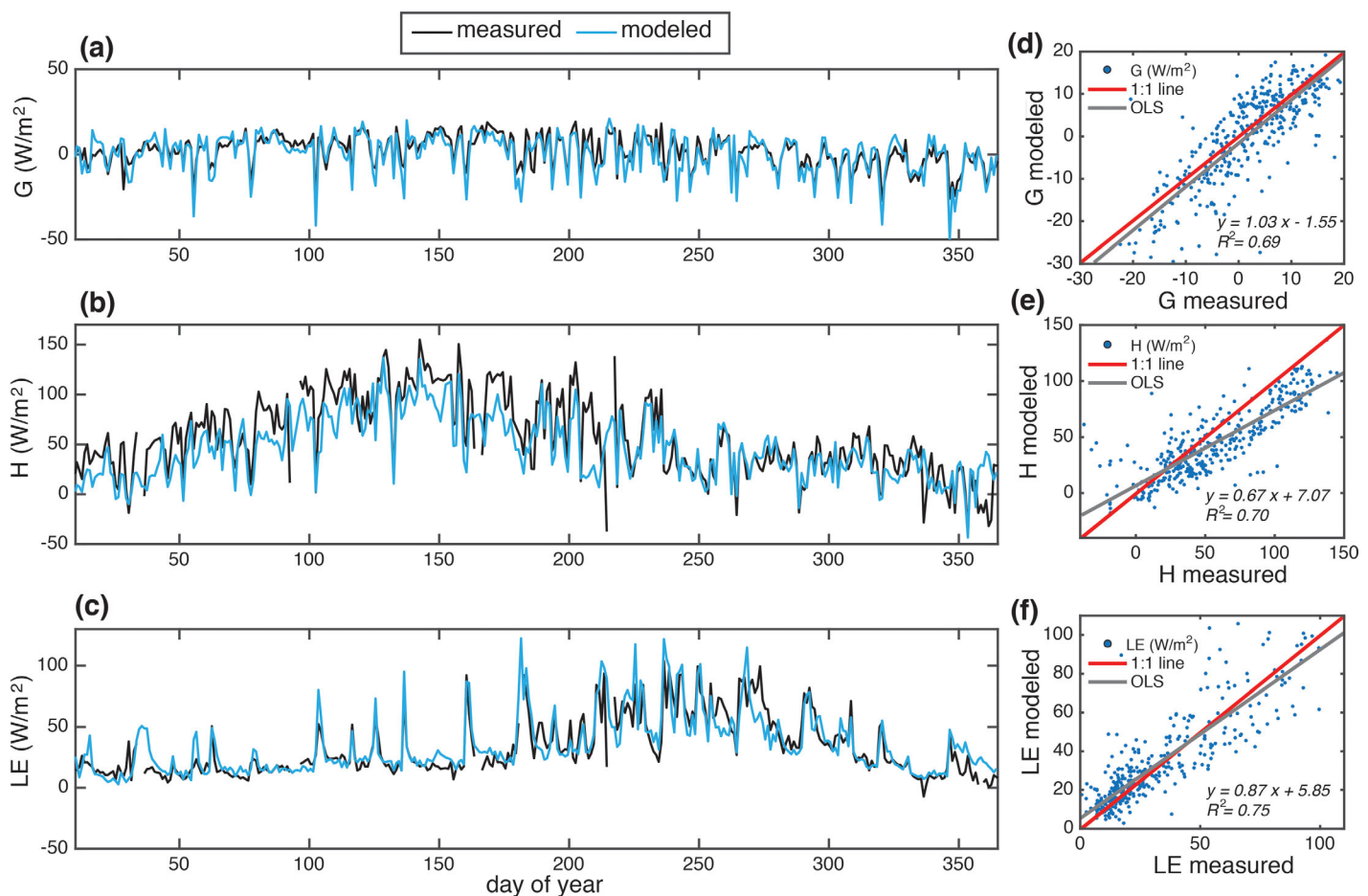


Figure 1. 2015 measured data of turbulent heat fluxes in Santa Rita Mesquite site used for model validation for daily averages of (a) ground heat flux G , (b) sensible heat flux H , and (c) latent heat flux LE . Scatter plots shown with a fitted regression line using ordinary least squares (OLS) in Figures 1d–1f. All regressions significant with $p < 0.01$. For further validation of the model refer to Figure A1 (in Appendix A) for residual plots of turbulent heat fluxes.

3.2. Measurements of Hydraulic Redistribution

To examine HR by mesquite roots, we used heat ratio method (HRM) to measure sap flux in lateral roots, taproot, and trunk of trees. Heat carried upstream and downstream is proportional to the magnitude of sap flux, which is an indicator of HR in the roots. Measurement of sap flux using SFM1 sap flow meter (ICT International, Australia) is a suitable approach since mesquites are ring porous and water transports through stem xylem in the outer tree ring (Scott et al., 2008). SFM1 measures sap flow with three 35 mm steel probes, with 5 mm spacing between them. Heat is emitted from the middle probe and the resulting heat flow is measured by thermistors at the tip of two neighboring probes. Positive measurements in the taproot indicate sap flux toward the mesquite through roots and negative measurements indicate sap flux away from the mesquite to the soil. The spacing of in situ sap flux sensors was calculated by using thermal diffusivity constant, median measurement time (80 s), and temperature increase from initial temperature at measurement probes (Burgess et al., 1998). Using the calculated probe spacing, we computed heat pulse velocity for probe misalignment. Mesquite wounding effect was corrected based on wound width and the sap flow meter installation (Barrett et al., 1995; Burgess et al., 2001a). For three mesquites, we monitored sap flux at trunk, taproot, and lateral roots. We excavated soil around roots and installed temperature probes at upstream and downstream of a pulsed heat source to measure heat transferred upstream and downstream, which is converted to sap flux (Burgess et al., 2001a, 2001b). After sap flux measurement probes are installed, excavated area was covered with plywood to protect exposed roots from daytime direct sunlight and nighttime freezing. Sap flux is monitored in 30 min intervals from May 2015 to May 2016. Barron-Gafford et al. (2017) provide further details on this methodology and the calculation of sap flux velocity.

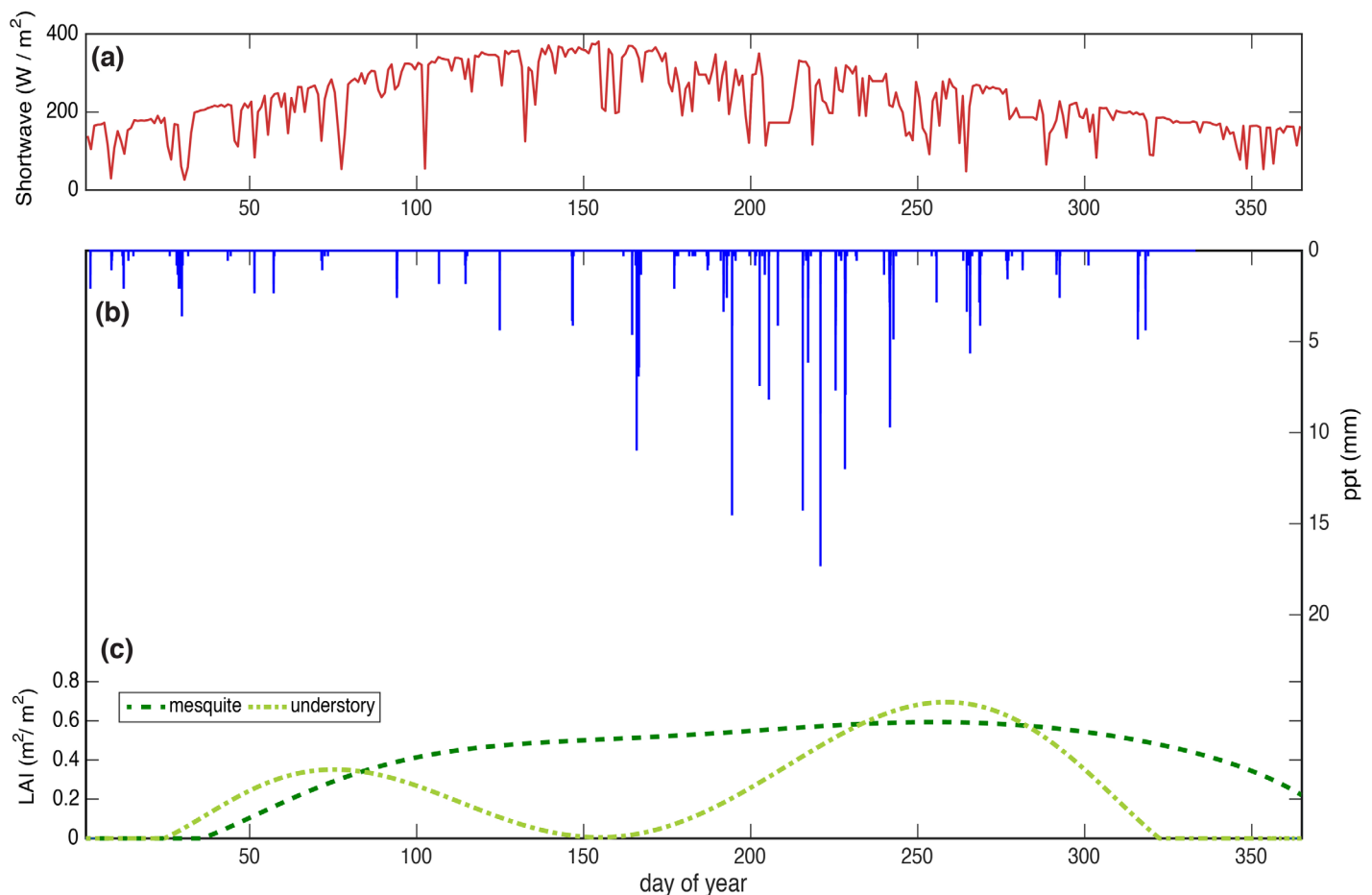


Figure 2. 2015 meteorological data from Santa Rita Mesquite site used for model simulation. (a) Daily averages of downward shortwave radiation, (b) precipitation (PPT), and (c) leaf area index (LAI) of overstory mesquite and understory bunchgrass.

3.3. Model Description

In an ecosystem, vegetation species coexist and interact with other species through above-ground processes such as radiation attenuation and energy partitioning, and below-ground processes, such as water and nutrient uptake. The multilayer canopy (MLCan) (Drewry et al., 2010a, 2010b; Le et al., 2011, 2012) model for the soil-root-plant continuum is a 1-D ecohydrological model, which is capable of simulating the interaction between multiple vegetation species, including plants with C_3 and C_4 photosynthetic pathways (Quijano et al., 2012). Importantly, it is therefore possible to quantify energy and water fluxes attributed to each vegetation species. For above-ground processes, the model uses the structure of multispecies vegetation to solve for partitioning of the solar radiation regime and water uptake pattern. For vertical leaf area density (LAD) calculation, LAD of each species is summed as a compound LAD, which determine radiation regime through vertical profile of coexisting vegetation species and the residual radiation reaching the soil surface. Then the energy is partitioned to different vegetation species according to the fraction of LAD of particular species that contributes to the compound LAD. Below-ground multispecies interaction in MLCan is accommodated through a shared resource model where the soil acts as a common reservoir allowing root access of multiple vegetation species to water and nutrient resources. Water use of multiple vegetation species in water-limited environment indicates whether the relationship between multiple species is competitive or facilitative. The model is also capable of simulating ecosystem with bare soil, by assuming that LAI equals to zero. To consider the fact that the study site is spatially heterogeneous, we run model simulations for area occupied with two vegetation species and for area between canopies. Then the two simulations are combined based on the fraction of intercanopy space over the area in different seasons.

The model computes stomatal conductance based on the Ball-Berry model (Ball et al., 1987). It describes the response of stomata conductance to the relative humidity, net photosynthesis, and CO_2 mole fraction.

Table 1
List of Parameters Used in the Simulation for Santa Rita Mesquite Site

Description	Symbol	Mesquite	Understory	Units
Photosynthesis				
Maximum rate of Rubisco carboxylation at 25°C ^a	$V_{c\ max25}$	17.62	39	$\mu\text{mol}/\text{m}^2\ \text{s}$
Maximum electron transport rate ^a	$J_{\max25}$	13.55		$\mu\text{mol}/\text{m}^2\ \text{s}$
Leaf respiration ^a	Rd_{25}	0.166		$\mu\text{mol}/\text{m}^2\ \text{s}$
Conductance and leaf physiology				
Ball-Berry slope ^b	m	9	12	
Ball-Berry intercept	b	0.01	0.008	$\text{mol}/\text{m}^2\ \text{s}$
Leaf potential at which half of the hydraulic conductance is lost ^c		−3		MPa
Leaf and canopy				
Leaf width (needle diameter) ^c	d_o	0.0312		m
Canopy height ^c	h_c	2.5		m
Flux tower observation height ^c	h_t	7.82		m
Soil property				
Percent of sand ^c		75		%
Percent of clay ^c		10		%
Root structure				
Maximum root depth	r_d	2.5		m
Fiftieth percentile rooting depth ^d	z_{50}	0.28	0.24	m
Ninety-fifth percentile rooting depth ^d	z_{95}	1.5	0.65	m
Litter Layer				
Litter layer depth ^a		0.03		m

^aMeasured in the study site. ^bBall (1988). ^cAmeriflux Network. ^dSchenk and Jackson (2002).

The slope (m) and intercept (b) of Ball-Berry equation depend on plant species and are obtained from literature (Ball, 1988). The intercept of Ball-Berry equation—the residual stomatal conductance when net photosynthesis is zero—denotes nighttime transpiration. Nighttime transpiration is detected when stomata does not close completely at night and water is lost due to vapor pressure difference between the leaf and atmosphere. Reduced canopy water potential due to nighttime transpiration can reduce HR (Howard et al., 2009). Since underestimating nighttime transpiration can result in overestimated hydraulic redistribution, obtaining values of residual stomatal conductance for modeling is crucial.

For below-ground water transport, the equation for HR is developed to simulate root-soil dynamics in the ecosystem (Mendel et al., 2002) and presented in Appendix B. The amount of hydraulically redistributed water is dependent on soil texture (Prieto et al., 2010). However, in this study, the study site has homogeneous loamy sand. For multispecies system, different vegetation species with differing rooting depth and root structure access different depths of soil and indirectly interact through the shared resources. Since soil is a common resource, soil moisture (θ) and soil water potential (ψ_s) are the same for all coexisting vegetation under the assumption that hydraulically redistributed water is well-mixed with the soil moisture, multiple species are linked through this shared resource. That is, a water release from one vegetation species may benefit another vegetation species that share the same soil layer.

The model is set up based on Amenu and Kumar (2008). Refer to Quijano et al. (2012, 2013) for multispecies modeling framework and the coupling of above-ground and below-ground water and energy dynamics. The details of MLCan development can be found in earlier publications (Drewry et al., 2010a) and the relevant parameters are included in Table 1.

The amount of the water used by each species for transpiration is calculated using *water use fraction* between two species, which is designated according to their *root fraction* and leaf area index. The contrasting characteristics of the deep-rooted mesquite and the shallow-rooted bunchgrass affect the depth and timing of soil moisture extraction. Soil moisture in the deeper depth is dominated by the overstory roots, whereas that in the shallower depth is shared between overstory and understory roots in proportion to their root mass distribution. The root fractions of overstory mesquite and understory bunchgrass are shown with the root distributions in different soil depths (Figure 3) and are computed as:

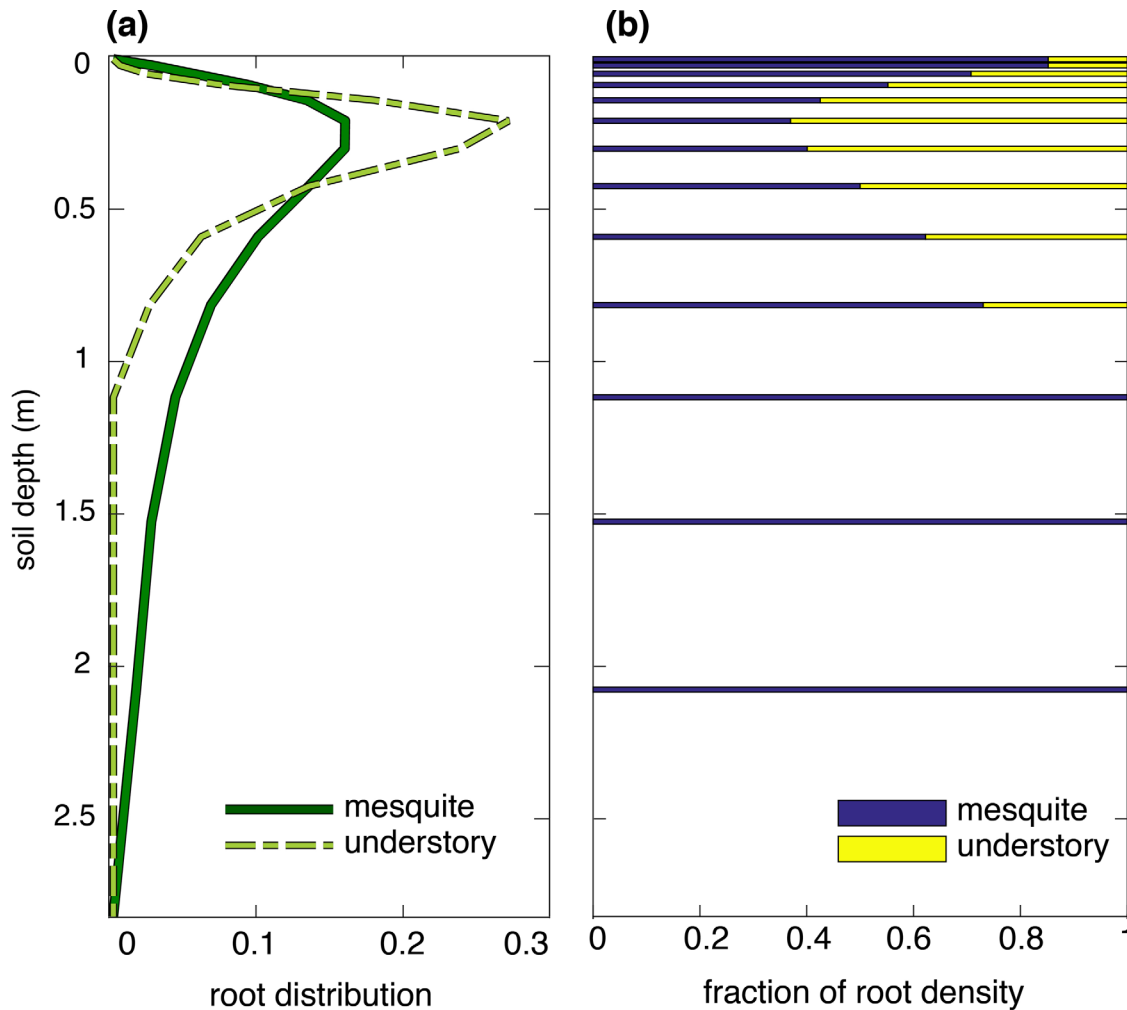


Figure 3. Illustration of root distributions and root fractions of overstory mesquite and understory bunchgrass. (a) Vertical root distributions of mesquite (dark green) and understory (light green) in different soil depths, and (b) root fractions of mesquite (horizontal navy bars) and bunchgrass (horizontal yellow bars) in according depths.

$$\text{Root fraction}_{\text{mesquite}} = \frac{F_{\text{root mesquite}}}{F_{\text{root mesquite}} + F_{\text{root understory}}} \quad (1)$$

$$\text{Root fraction}_{\text{understory}} = \frac{F_{\text{root understory}}}{F_{\text{root mesquite}} + F_{\text{root understory}}}$$

Vertical root distribution ($F_{\text{root mesquite}}$ and $F_{\text{root understory}}$) is calculated according to equation (B2) in Appendix B (Schenk & Jackson, 2002).

Leaf areas of the two species are also included in the calculation of water use fraction. Overstory mesquite maintains high leaf area all year long except for the dry winter period, whereas understory bunchgrass greens up only during spring (given sufficient seasonal precipitation) and summer monsoon season (Figure 2). Therefore, mesquite uses soil moisture through most of the year, whereas bunchgrass uses it between dormant periods. Therefore, when leaf area of understory remains low during dry season, water is mostly used by overstory mesquite and water is shared between two species as understory bunchgrass grows. We assume that the water use fractions of mesquite and understory are determined by the combined effect of their root and LAI fractions, obtained as:

$$\text{LAI fraction}_{\text{mesquite}} = \frac{\text{LAI}_{\text{mesquite}}}{\text{LAI}_{\text{mesquite}} + \text{LAI}_{\text{understory}}} \quad (2)$$

$$\text{LAI fraction}_{\text{understory}} = \frac{\text{LAI}_{\text{understory}}}{\text{LAI}_{\text{mesquite}} + \text{LAI}_{\text{understory}}}$$

To calculate a fraction that combines both root fraction and LAI fraction, we multiply root fraction and LAI fraction for each vegetation species and obtain *root LAI fraction*, as:

$$\text{Root LAI fraction}_{\text{mesquite}} = \text{Root fraction}_{\text{mesquite}} * \text{LAI fraction}_{\text{mesquite}} \quad (3)$$

$$\text{Root LAI fraction}_{\text{understory}} = \text{Root fraction}_{\text{understory}} * \text{LAI fraction}_{\text{understory}}$$

Then we calculate water use fraction of each vegetation species by dividing each root LAI fraction by the sum of root LAI fractions of both species. Hence, we get the water use fraction that quantifies the soil water access by trees and grasses in different soil depths throughout the year in 30 min intervals, as:

$$\text{Water use fraction}_{\text{mesquite}} = \frac{\text{Root LAI fraction}_{\text{mesquite}}}{\text{Root LAI fraction}_{\text{mesquite}} + \text{Root LAI fraction}_{\text{understory}}} \quad (4)$$

$$\text{Water use fraction}_{\text{understory}} = \frac{\text{Root LAI fraction}_{\text{understory}}}{\text{Root LAI fraction}_{\text{mesquite}} + \text{Root LAI fraction}_{\text{understory}}}$$

The model simulation provides information about moisture transport between soil and roots for each soil layers. Based on the direction of water transport, we identified water uptake and release as water moving from soil to root and vice-versa. When water is absorbed by roots in deep layers and released in shallow layers, we classified it as HL, and when water is absorbed in shallow layers and released in deep layers, we classified it as HD. This classification is done for each 30 min model time step.

4. Results and Discussion

4.1. Seasonality of HR

Figures 4 and 5 show the HR response to a precipitation pulse that occurred in a dry and wet periods, respectively. Figure 4c shows that HL occurs most often in the hot and dry month of May (around DOY 152) when water at night is transported to shallow soil depths and released into the soil, and during the day water is extracted throughout most of the soil profile to support mesquite transpiration. Figure 4 also shows that the precipitation event (DOY 159–160) switches the root water transport from HL to HD. Figure 5c shows HD in August during a summer monsoon period (around DOY 222). During wet period, in the daytime, roots take water from the near-surface soil whereas in the nighttime, roots take water from shallow soil layers and release water in deep soil layers.

Figure 6 shows different components of moisture transport processes in a time period when HL switches to HD due to a precipitation event. Before the precipitation on DOY 236, both measured sap flow (in panel (e)) and modeled HR (in panel (f)) are positive, which means that sap flow is upward along the taproot and there is hydraulic lift. After the precipitation, sap flow at night and modeled HR become negative. While not directly comparable, both panels (e) and (f) show an overall agreement between measured sap flow velocity in a taproot and hydraulic redistribution patterns. However, the model-generated result shows that water is hydraulically descended during both the daytime and nighttime (Figure 6c and 6f), whereas downward flow does not occur during the daytime in sap flow measurements (Figure 6e). During this wet period, the model shows that there is sufficient water in the near-surface layers to meet the evapotranspiration demand and contribute to HD even during the day. This appears to be slightly inconsistent with the observational results, which shows upward flow during the daytime. This could be an indicator of a mismatch between model estimate of overall moisture transport through the vegetation system, and sap flow measurement in the taproot, which is obtained at around 50 cm below the surface. At this depth of measurement, upward sap flow may also result in release in the soil through the lateral roots that lie above the 50 cm depth, but there is no way to ascertain that this is the case. Around DOY 237, we observe that the amount of released water decreases when water demand from transpiration peaks in daytime. This daytime HR phenomena may also occur on a clear day when critical xylem water potential is higher than soil water potential of the rooting zone (Hultine et al., 2003). In Figure 6, we observe daytime HR with high mesquite transpiration, which suggests that daytime water release by roots may be due to the moisture gradient from root to surrounding soil. The depth where the water is released ranges from the shallow soil layers

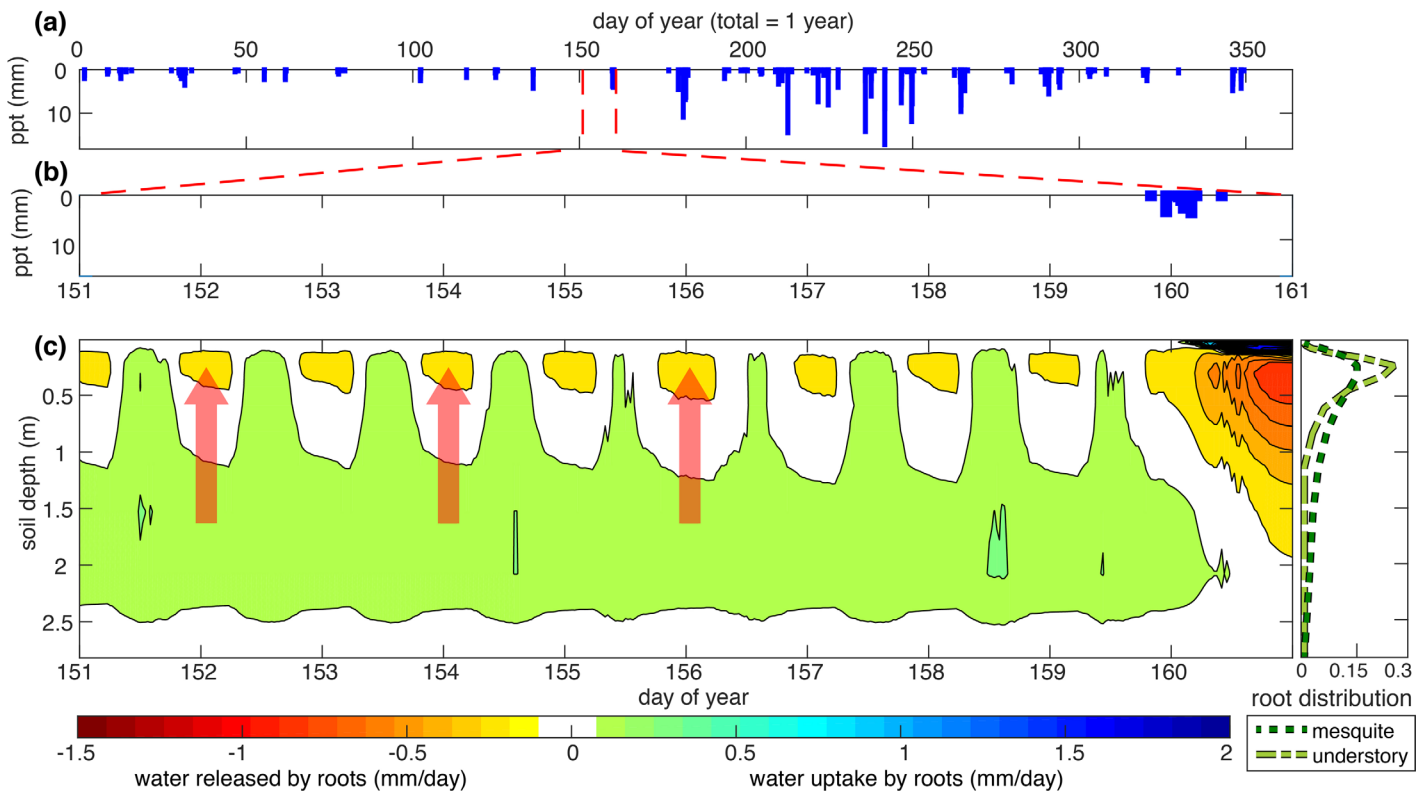


Figure 4. Illustration of model-generated hydraulic lift during a dry period. (a) 2015 precipitation, (b) precipitation within a selected 10 day window, and (c) depth and magnitude of water uptake and release by roots for the 10 day period (red arrows represent the direction of vertical water flow), with mesquite and understory vertical root density distribution shown as dotted lines for mesquite (dark green) and understory bunchgrass (light green).

to deep soil layers, indicating that the release during the day is caused by large soil water potential gradients.

Note that Figure 6e plots the sap flow velocity in a mesquite taproot and Figure 6f plots water uptake (or release) at soil-root interfaces. Therefore, values plotted in panels (e) and (f) have orders of magnitude difference. In order to compare measured sap flow and simulated water transport, we convert sap flow velocity and water transport rate into volumetric measures. Measured taproot sap velocity is multiplied by total taproot sapwood area of 95.6 cm^2 to obtain volume of sap flow moving across the total sapwood area (Figure A3a). Based on Scott et al. (2008), measurements of root diameter and SWA are plotted in Figure A3b in Appendix A. The volume of water transport in the model is calculated using a mesquite canopy area of 22.0 m^2 from the measured tree. Volume of measured sap flow and volume of simulated water transport (hydraulic redistribution plus canopy transpiration) are plotted in Figure 7b. Blue line in the figure illustrates volume of sap flow along a taproot, where positive direction accounts for upward movements due to both hydraulic lift and transpiration. Overall, the sum of simulated HR and transpiration agree with the sap flow in terms of patterns and magnitudes. However, for upward flow, model-generated water transport volume tends to be larger than measured sap volume.

Figure 7c shows model-generated HR, classified into HL and HD. As we hypothesized for systems without deep soil water source, we found that HL occurs more episodically and HD dominates through this period. We see that a large rain event creates water potential gradient along soil depths and as a result, a large HR flux is generated from shallow soil layers to deeper soil layers through roots. Therefore, water is transported downward through mesquite roots (HD), potentially reducing water loss due to surface evaporation. Significant HD during summer monsoon (Figure 7c) suggests that HR may extend the growing season of mesquites when surface soil dries after the monsoon, thus supporting transpiration through seasonal dry periods (Scott et al., 2008). Also, HD may reduce water availability for shallow-rooted understory plants. HL occurs between precipitation events when water supply from precipitation is absent and it increases water availability in shallow layer. Understory plants may benefit from water resources supplied by deep-rooted trees and survive longer during dry periods.

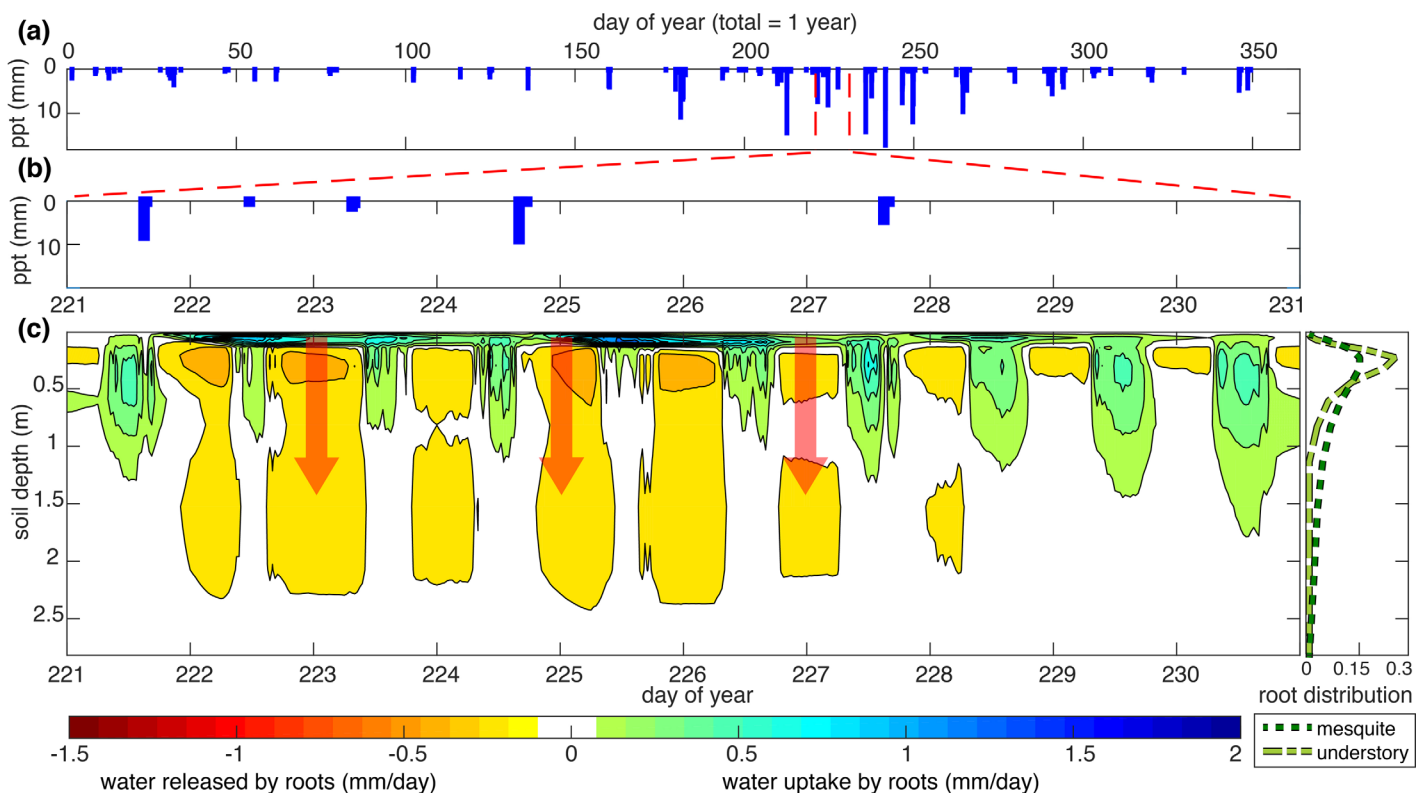


Figure 5. Same as Figure 4 but for hydraulic descent during a wet period. (a) 2015 precipitation, (b) precipitation within a 10 day window, and (c) depth and magnitude of water uptake and release by roots for a selected 10 day period (red arrows represent the direction of vertical water flow), with mesquite and understory vertical root density distribution shown as dotted lines for mesquite (dark green) and understory bunchgrass (light green).

4.2. Impact of HR on Transpiration Water Use

Water dynamics in natural ecosystems is an important determinant of above-ground and below-ground interactions between coexisting vegetation. In this section, we examine the role of HR on water dynamics in a semiarid savanna ecosystem with two species vegetation. Table 2 presents the 2015 water budget and its components, and Figure 8 shows the relative partitioning in terms of the annual precipitation. Total transpiration (T) use in 2015 is 232 mm, with slightly more water used by the mesquite, and total evapotranspiration (ET) is 383 mm. This results in a T/ET ratio of about 61%, which is nearly equal to the 62% determined for a long-term average at this site (Scott et al., 2015). Vegetation evaporation is only 11 mm, which refers to the water evaporating from canopy and leaf interception. In 2015, 474 mm of water is supplied to the system as precipitation and plants moved 64 mm of water—13% of total precipitation, which equates to 28% of the moisture not lost through transpiration—to deeper soil layers through roots (HD). This is a surprising amount of stored water as the amount of storage changes in the vadose zone is typically small (Scott, 2010). This stored water was likely due to the unusually large amount of precipitation that fell toward the end of the monsoon season. Annually, 8% of HD water is redistributed up to shallow soil layers and the rest of HD water either increases soil moisture storage in the deeper soil layers or is consumed by mesquites for transpiration. In this site, the amount of hydraulically lifted water is considerably smaller than that of hydraulically descended water because there is no connectivity to groundwater or access to a deep moisture source (David et al., 2013, Fu et al., 2016). The comparison of the magnitude of HL and HD indicates that there is a net downward movement of water through roots, resulting in a negative HRR ($HD > HL$), which may inhibit understory growth or may support a more temporally episodic and spatially intermittent pattern.

Partitioning of the transpiration between overstory mesquite and understory bunchgrass indicates competitive relationship between the species in the ecosystem. We use the calculation of water use fraction to attribute the water in shallow and deep soil layers to shallow-rooted bunchgrass and deep-rooted mesquite

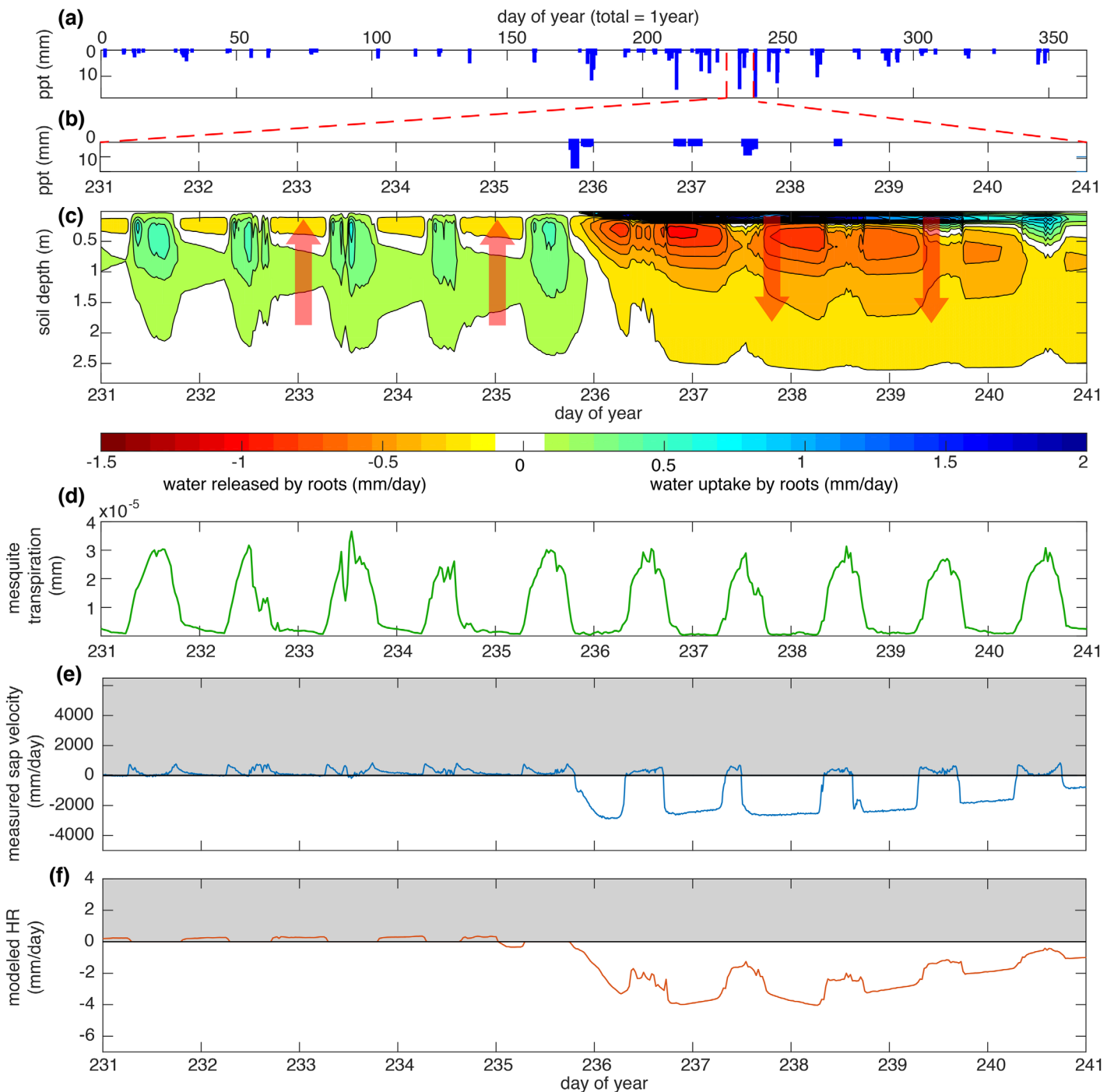


Figure 6. Illustration of model-generated hydraulic redistribution shift from hydraulic lift to hydraulic descent during a wet period. (a) 2015 precipitation, (b) precipitation in a selected 10 day window, and (c) depth and magnitude of water uptake and release by roots for the period (red arrows represent the direction of vertical water flow). Note that positive values are water uptake by roots and negative values are water release from root to soil. (d) Modeled mesquite transpiration, (e) measured taproot sap flow velocity, and (f) amount of water released from roots to soil at all depths in each time step, which we denote as modeled HR. In Figure 6e, positive and negative values represent the upward and downward flows, respectively, through the roots. In Figure 6f, positive values represent the HL and negative values represent the HD.

(equation (4)). Assuming HR water in different soil layers is thus attributed to bunchgrass and mesquite for their transpirations, HR supports 47% of mesquite transpiration. Annually, HR has beneficial effects more on mesquite transpiration than on bunchgrass transpiration. In this study site, the relationship between

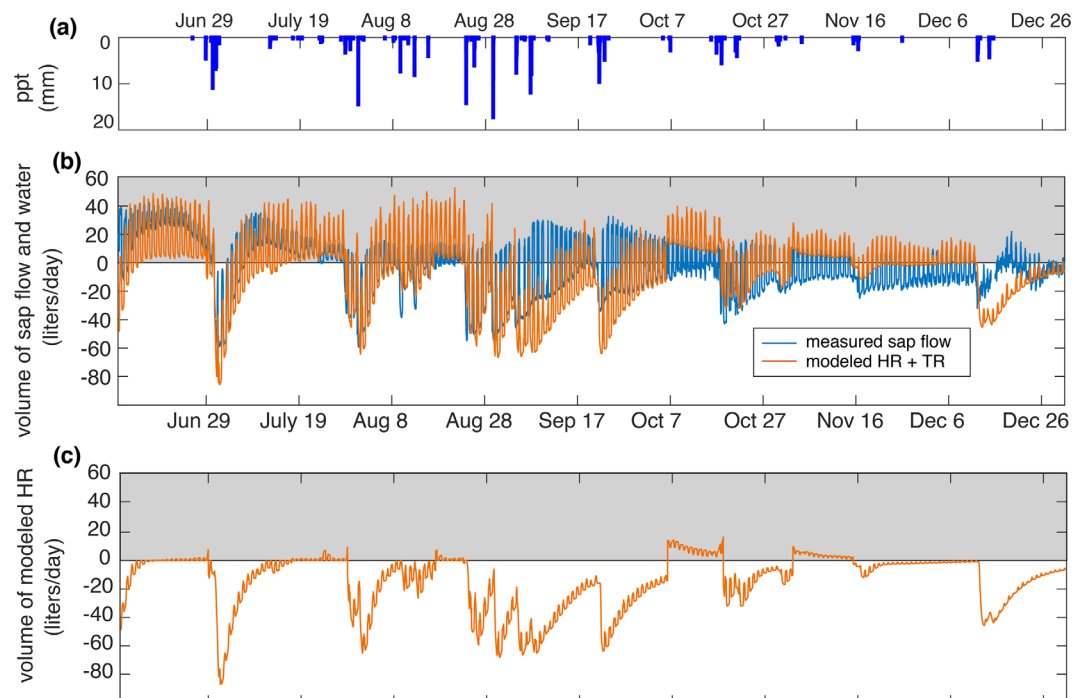


Figure 7. Comparison of measured volume of sap flow and modeled volume of HR water. (a) 2015 precipitation and (b) volume of sap flow measured at the mesquite taproot from the field (blue) and volume of water from HR and transpiration (HR + TR, orange). Positive values represent the upward flow and negative values represent the downward flow. (c) Volume of modeled HR, where positive values represent the HL and negative values represent the HD.

coexisting vegetation is more competitive than facilitative as shown in the result using evaporative water use fraction (Figure 8, inner ring).

Figure 9 shows a precipitation pulse analysis of ecosystem evapotranspiration and the water sources that supply the evapotranspiration demand. Water used for transpiration is supplied from either hydraulically redistributed water or infiltrated soil water. During the daytime, both overstory mesquite and understory bunchgrass use water from HR and soil moisture for transpiration. During nighttime, when the amount of water released from roots is larger than the amount of water absorbed by roots, the water is stored in soil. Precipitation from DOY 236 to 238 supplies water to be redistributed by both mesquite and bunchgrass. For mesquite, the amount of HR water available is larger than the amount of water used in evapotranspiration. Therefore, water is stored for more than 8 consecutive days (until DOY 245) in the soil depths where mesquite can access. The amount of HR water available for bunchgrass is smaller compared to that for mesquite. The water available for bunchgrass is stored for only 2 or 3 days. For the same precipitation event, the time period that a mesquite benefits from HR is longer than the period that bunchgrass benefits. We

Table 2
Amount of Water Annually Used in Different Parts of Water Dynamics in Santa Rita Mesquite Site in 2015

	Total	Overstory mesquite	Understory bunchgrass
PPT	474 (mm)		
Vegetation Transpiration	232 (mm)	128	104
Soil evaporation	151 (mm)		
Soil moisture	80 (mm)		
Vegetation evaporation	11 (mm)	8	3
HD	64 (mm)	56	8
HL	5 (mm)	4	1
HD/PPT	13 (%)		
HL/HD	8 (%)		

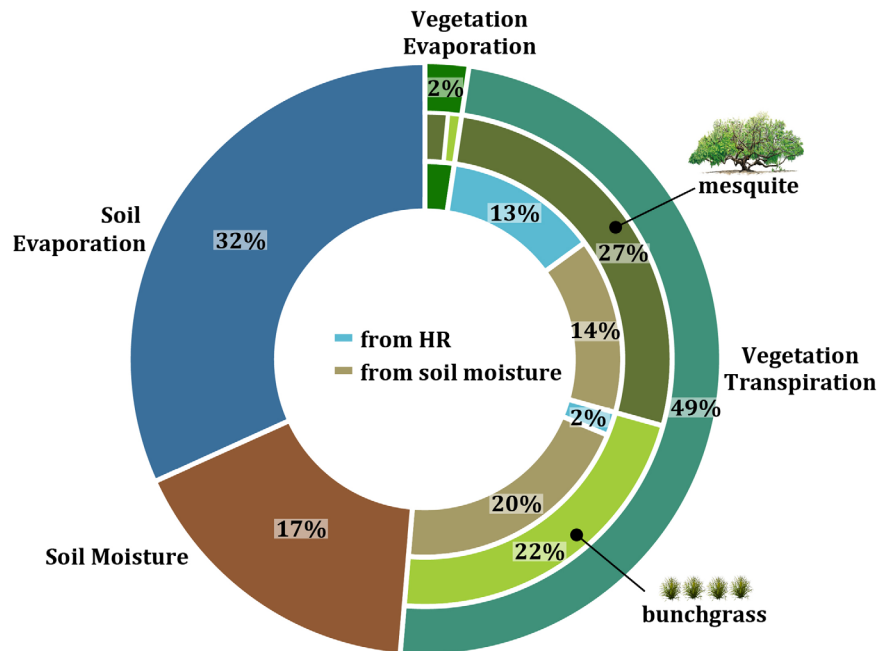


Figure 8. Illustration of the partitioning and fate of annual precipitation at the Santa Rita site in 2015. (Outer ring) Partitioning of water supplied to the ecosystem by precipitation, (middle ring) partition of vegetation transpiration between the overstory mesquite and understory bunchgrass, and (inner ring) water source partitioning from HR and soil moisture supporting mesquite and bunchgrass transpiration (based on data in Table 2).

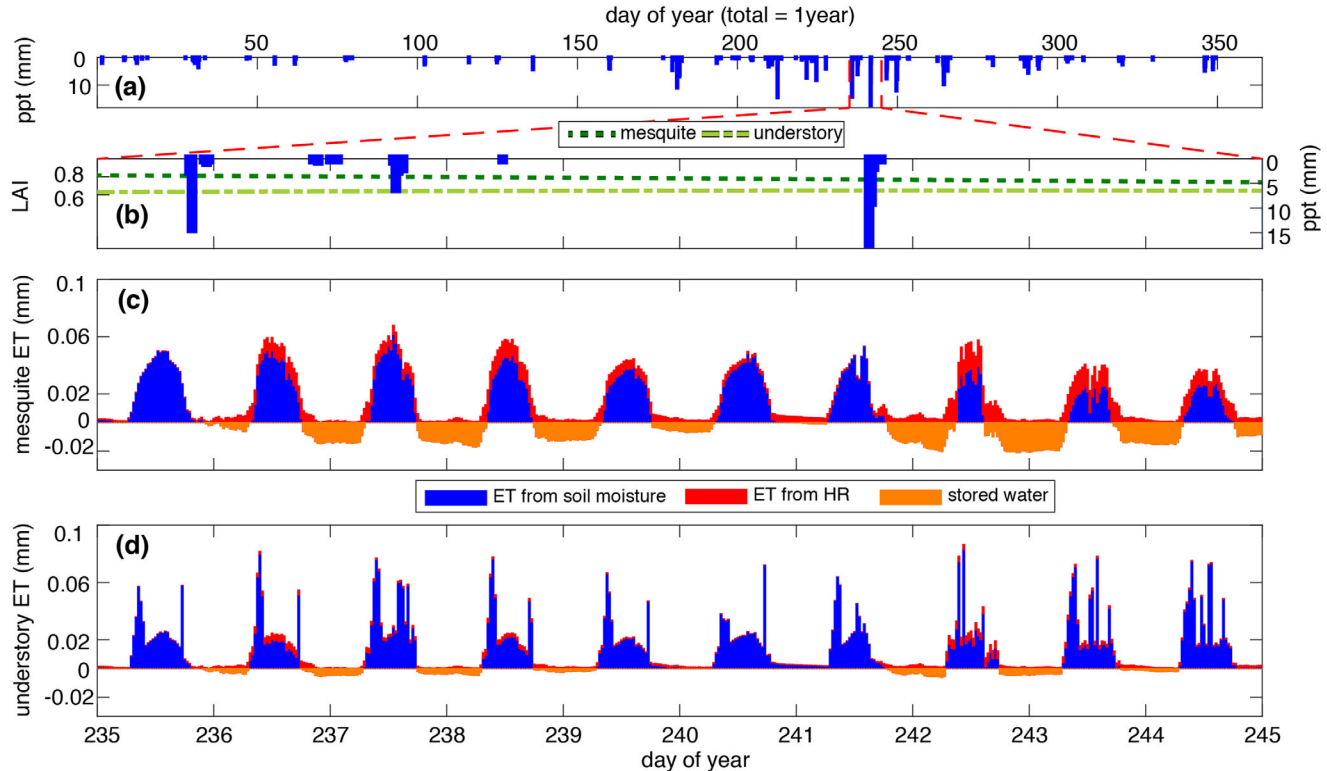


Figure 9. Illustration of the water sources supporting evapotranspiration. (a) 2015 precipitation, (b) precipitation with mesquite and understory LAI for a 10 day period during monsoon, (c) overstory mesquite evapotranspiration from soil water that has been released by HR (red) and by infiltration (blue), and (d) understory bunchgrass evapotranspiration from HR (red) and soil water (blue). When the amount of water released from roots is larger than amount of water absorbed by roots, water is stored in soil (orange).

calculated the number of days that water is redistributed after a precipitation event for all of 2015. For a pulse of precipitation, on average the water is stored for about 9 days (averaged over 14 precipitation events) in the soil layers where mesquite can use it, whereas the water is stored for around 4 days in the soil layers where bunchgrass can access it.

5. Summary and Conclusions

This study explored how the movement of soil moisture by overstory vegetation through hydraulic redistribution may impact near-surface available moisture and water utilization by coexisting overstory mesquite and understory bunchgrass in a semiarid savanna. We examined how the presence of hydraulic redistribution prompts competitive dependencies between coexisting vegetation in water-limited environment. This is in contrast to some other ecosystems, where hydraulic redistribution may stimulate both competitive and facilitative dependencies between coexisting species (Quijano et al., 2012). In this study, integration of numerical modeling with field measurements provided insights into the hydrologic mechanisms that drive the dynamic water movement. We identified the hydraulic redistribution regime of the ecosystem to capture the dynamic heterogeneity of soil moisture availability that arises due to water uptake patterns of plant roots. Exploring hydraulic redistribution regime of other ecosystems may provide insights to ecohydrological patterns of water and carbon cycling facilitated by soil moisture transport through plant roots.

We tracked the water partitioning in the ecosystem with two vegetation species using the shared resource modeling approach to determine the relationship between coexisting vegetation and to quantify the use of the limited amount of water between overstory and understory. The key findings from the study are:

1. In 2015, 83% of the precipitation water entering the ecosystem is transpired through vegetation or evaporated from the soil. Mesquite and bunchgrass transpired 27% and 22% of this water, respectively. Water loss due to soil and canopy evaporation is 34% and the rest of the water is stored in soil that is used in the following dry season.
2. Annually, 13% of precipitation is transferred to deep soil layer through HD. During the wet season, mesquites moved water to deeper soil through their roots right after rain, potentially preventing water loss due to surface evaporation. 8% of this HD water is transported back to shallow soil layers through HL during the dry season. We find that the deeper-rooted mesquite benefited from water supplied by both HD and HL and shallow-rooted bunchgrass had access to only HL water. HR annually supports 47% of mesquite transpiration. However, the impact of HR on bunchgrass transpiration is less significant (9%). In this water-limited environment, HR provides a competitive advantage for water use to mesquite over bunchgrass.

Therefore, we conclude that the overall net downward movement of water through roots to deeper soil layers in this savanna illustrates that the mesquites are the primary utilizers of the water, which provides a competitive advantage to overstory mesquite over understory bunchgrass.

Appendix A: Validation of MLCan Model

In a synthesis of experimental and modeling work, model validation is a crucial part. To validate the model, we provide discussion of residuals, comparison of soil moisture, and configuration of modeled HR validation.

Validation of the modeled turbulent heat fluxes is presented in section 3.1. For a further validation of the model, residual plots of the heat fluxes are plotted in Figure A1. Residuals are calculated by subtracting modeled heat fluxes from measured heat fluxes. Residuals cluster along the zero line in Figures A1a–A1c and data points follow the standard normal line in Figures A1d–A1f, indicating an unbiased set. For ground heat flux, residual values deviate more from zero line for low values of modeled G (Figure A1a), indicating that low modeled ground heat fluxes are underestimated. On the other hand, for latent heat flux, residual values are negative and deviate more from the zero line for high values (Figure A1c). In Figure A1b, residuals of sensible heat scatter along a line slightly higher than zero, indicating that the model generally underestimates sensible heat flux.

To validate the simulated result of below-ground processes, we compare measured and modeled soil moisture at different depths. Overall, modeled soil moisture matches with measured soil moisture at all soil

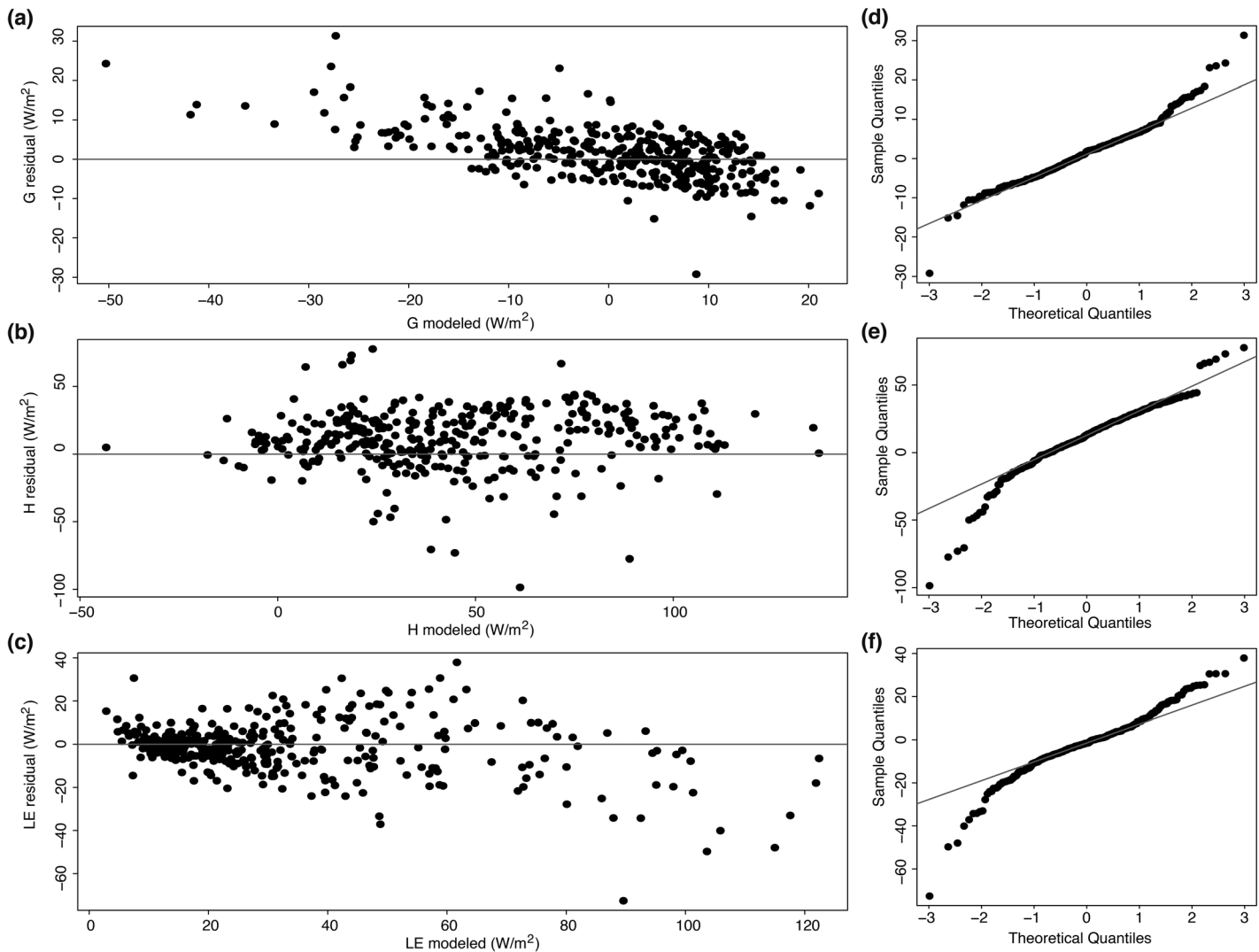


Figure A1. Residual (difference between measured and modeled heat fluxes) plots of turbulent heat fluxes in Santa Rita Mesquite site for daily averages of (a) ground heat flux, (b) sensible heat flux, and (c) latent heat flux. Quantile-quantile plots are shown in Figures A1d–A1f.

depths. High peaks in both measured and simulated soil moisture at shallow depths are generated as a response to precipitation (Figures A2a–A2c).

Before conducting a comparison between measured data and simulated results, we must convert the values in comparable measures. Before we compare taproot sap flow velocity and water transport rate at soil-root interfaces, we must convert both rates to volumetric measures as described in section 4.1. Cross section areas, sapwood area (SWA, cm^2) and canopy area (m^2), used for the conversion are illustrated in Figure A3. SWA—the difference between taproot area and heartwood area—is a cross section area through which the sap is transported (Figure A3b). Mesquite canopy area is a cross section area through which the water transpires from a mesquite. Using SWA and canopy area, we convert sap flow velocity and water transport rate to volumetric measures that are compared for validation of modeled HR.

Appendix B: Description of MLCan Model

The multilayer canopy (MLCan) model for soil-root-plant continuum is a 1-D ecohydrological model, which is used to simulate above-ground and below-ground linkages. MLCan couples leaf-level ecophysiological processes with physical processes and below-ground moisture transport using detailed incorporation of

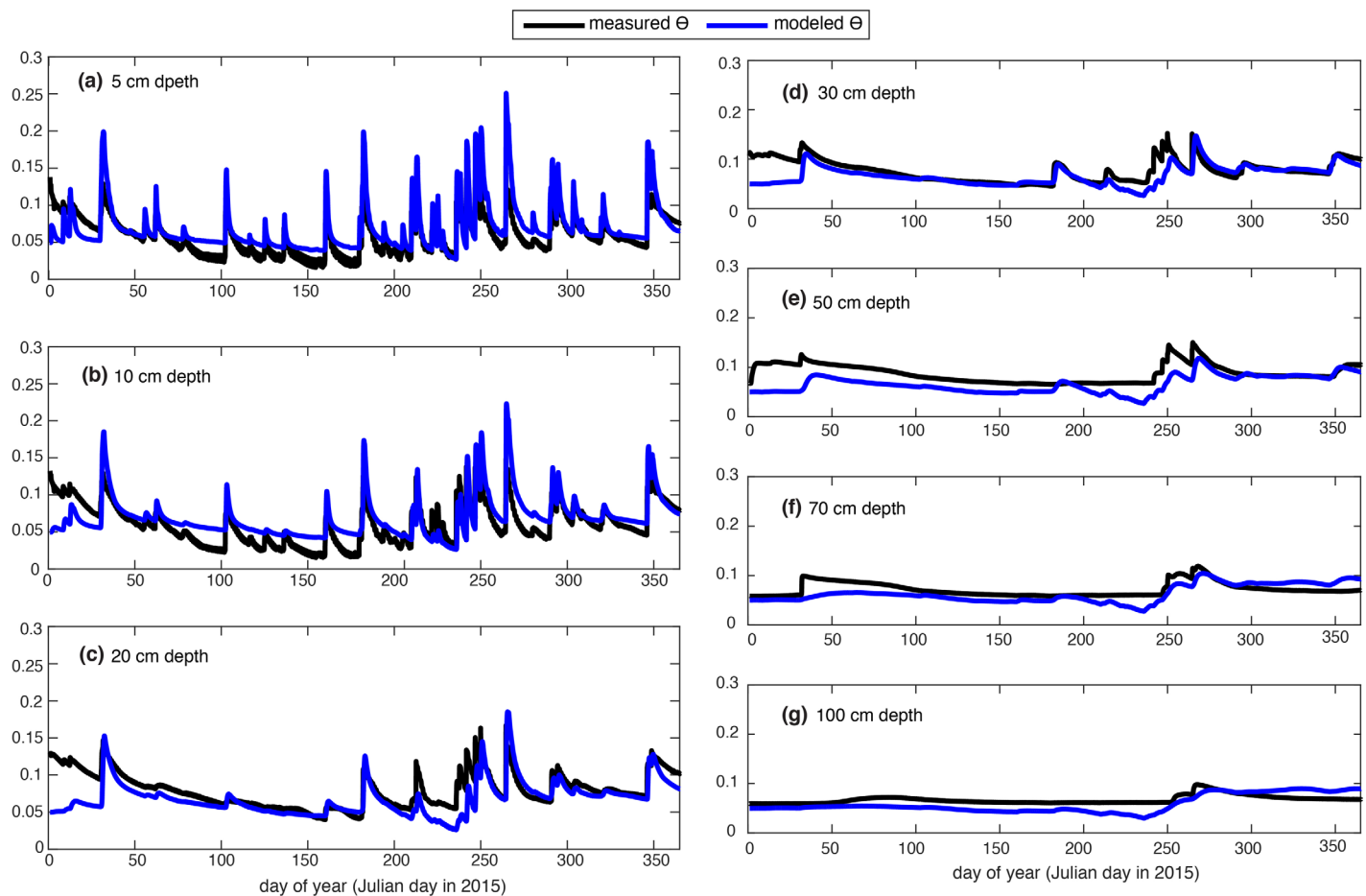


Figure A2. 2015 measured soil moisture data (black) at the Santa Rita Mesquite site compared with modeled soil moisture (blue) in different depths: (a) 5 cm, (b) 10 cm, (c) 20 cm, (d) 30 cm, (e) 50 cm, (f) 70 cm, and (g) 100 cm.

photosynthesis, stomatal conductance, energy balance, and boundary layer conductance. The model resolves shortwave and longwave radiation regimes through multiple layers of vegetation canopy by leaf area density profile (Drewry et al., 2010a). The model takes inputs such as shortwave radiation, longwave radiation, wind speed, air temperature, atmospheric vapor pressure, and precipitation. As a solution of leaf energy balance, the model generates latent and sensible heat fluxes and CO_2 assimilation for different canopy layers. Based on the vertical leaf area density structure, forcing variables are turned into vertical profile. Canopy model solution starts with solving the within canopy radiation regimes. Direct shortwave radiation at the top of the canopy is calculated according to Beer-Bouguer-Lambert's law and diffused shortwave radiation is also calculated through different canopy layers. Longwave radiation, which is absorbed, reflected, transmitted, and emitted, is also calculated. The residual radiation after shortwave and longwave calculation is set to reach soil surface directly. After solving for radiation, energy and water fluxes for soil are calculated. For soil moisture, Richards' equation coupled to moisture transport equation through the root system (Amenu & Kumar, 2008) are solved and vertical soil moisture profile is calculated at each time step along with hydraulic redistribution. For leaf photosynthesis, refer to Farquhar equation (Farquhar et al., 1980) and for stomatal conductance, refer to Ball-Berry equation (Ball et al., 1987). Leaf energy balance is calculated using Nikolov (1995) and soil energy balance is determined using the numerical solution as provided in Oleson et al. (2010).

Equation for HR is provided below. The coupling between root and soil is solved for all vegetation species simultaneously. The model simulates water flow through roots with steady state approximation coupled with unsteady Richards equation, an equation commonly used to examine water flow in soil (Amenu & Kumar 2008; Quijano et al., 2012). The equations used for below-ground soil moisture are provided below.

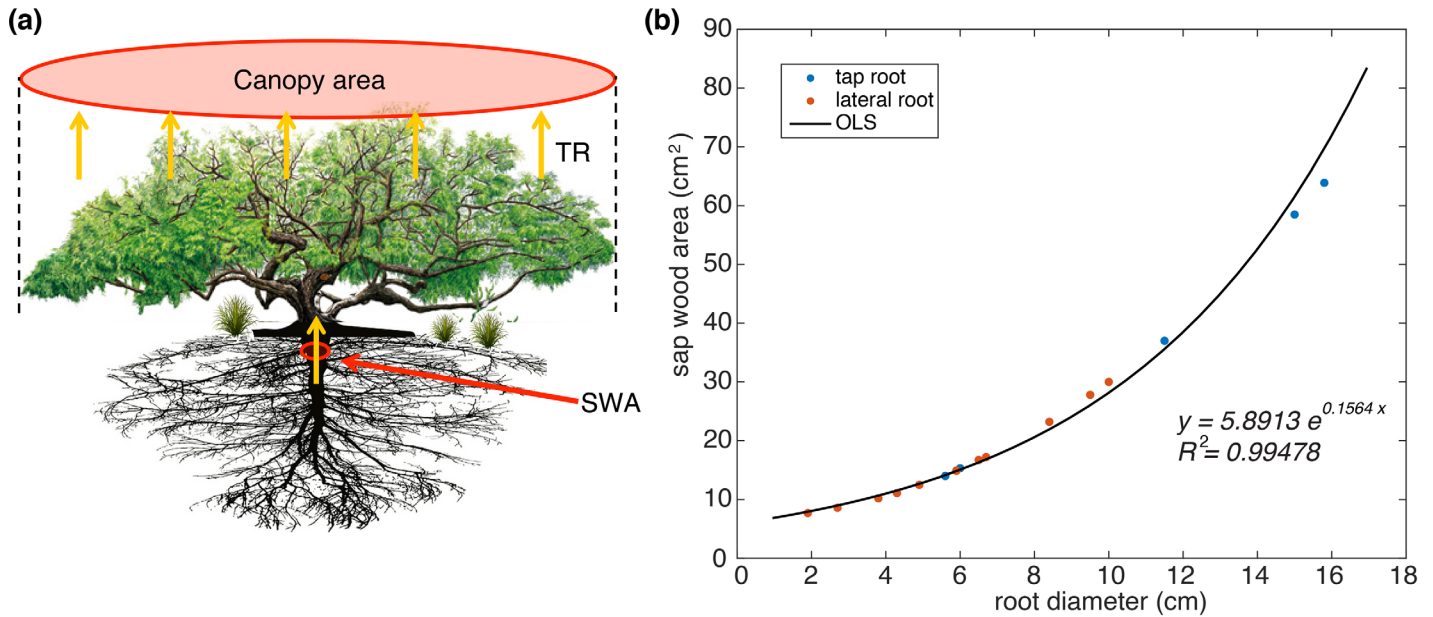


Figure A3. (a) Illustration of cross section areas of sap flow and transpiration (TR) used for converting rates to volumetric measures. Sap flow velocity and transpiration rate are multiplied by taproot sapwood area and canopy area, respectively, resulting in volumetric amount of water. (b) Diameters and SWAs of the mesquite lateral and tap roots (≥ 2 cm diameter) from Scott et al. (2008), used to determine total taproot sapwood area.

HR flux occurs from plant root to the surrounding soil when root water potential (ψ_r) becomes higher than soil water potential (ψ_s). For cases where HR is suppressed, radial root hydraulic conductivity is set equal to zero when root water potential is higher than soil water potential

$$\begin{aligned} \frac{\partial \theta}{\partial t} - \frac{\partial}{\partial z} \left[K_s \left(\frac{\partial \psi_s}{\partial z} - 1 \right) \right] &= - \sum_{i=1}^M K_{r_i}^R (\psi_s - \psi_{r_i}) \\ - \frac{\partial}{\partial z} \left[K_{r_1}^A \left(\frac{\partial \psi_{r_1}}{\partial z} - 1 \right) \right] &= K_{r_1}^R (\psi_s - \psi_{r_1}) \\ - \frac{\partial}{\partial z} \left[K_{r_2}^A \left(\frac{\partial \psi_{r_2}}{\partial z} - 1 \right) \right] &= K_{r_2}^R (\psi_s - \psi_{r_2}) \\ &\vdots \\ - \frac{\partial}{\partial z} \left[K_{r_M}^A \left(\frac{\partial \psi_{r_M}}{\partial z} - 1 \right) \right] &= K_{r_M}^R (\psi_s - \psi_{r_M}) \end{aligned} \quad (B1)$$

In equation (B1), M is number of vegetation species, ψ_s is water potential in the soil, and ψ_r is water potential in the roots of i th vegetation species. The term θ represents the soil moisture and z represents the vertical coordinate and t represents the time step. The term K_s is hydraulic conductivity of soil, and $K_{r_i}^R$ and $K_{r_i}^A$ represents the radial and axial root conductivities of i th vegetation species.

One of the main factors that determine HR is plant root biomass distribution in the vertical direction. Root biomass distribution is a unique characteristic related to vegetation species and climate zone. In the model, root biomass distribution is generated based on published literature values, and presented in equation below (Schenk & Jackson, 2002):

$$F_{root}(z) = \left[\frac{1}{1 + \left(\frac{z}{z_{50}} \right)^c} \right] \quad (B2)$$

Parameters for calculating vertical root distributions in the soil profile are also extracted from the same literature. In the equation, $F_{root}(z)$ represents the cumulative fraction of root biomass above a depth of z and z_{50}

is the depth where half of cumulative root biomass is located, and z_{95} is the depth where 95% of cumulative root biomass is located. The term c is a dimensionless shape parameter, calculated by dividing a constant value by the difference between $\log z_{50}$ and $\log z_{95}$ (Schenk & Jackson, 2002). Values of MLCan parameters for the study site, including z_{50} and z_{95} , are presented in Table 1.

Acknowledgments

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